



3 Complex Analysis 2024

Additional Problems

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Problem 1 Determine the set of all $z \in \mathbb{C}$ where the function $f(z) = |z|^{\frac{3}{2}}$ is differentiable.

Problem 2 Let f be a holomorphic function in a disc $D(z_0; R)$ with $z_0 \in \mathbb{C}$ and $R > 0$. Prove that if $\text{Im } f(z)$ is constant in $D(z_0; R)$, then $f(z)$ is constant in $D(z_0; R)$.

Problem 3 Determine the radii of convergence of the following power series:

(a) $\sum_{n=0}^{\infty} (-2)^n (z + i)^n,$

(b) $\sum_{n=0}^{\infty} \frac{2^n}{(2n)!} z^n,$

(c) $\sum_{n=0}^{\infty} \frac{(n!)^2}{(2n)!} z^{2n},$

(d) $\sum_{n=2}^{\infty} 2^{\ln n} z^n,$

(e) $\sum_{n=0}^{\infty} \frac{\sqrt{(2n)!}}{n!} z^{2n},$

(f) $\sum_{n=1}^{\infty} \left(\frac{n^2}{(n+1)(2n+1)} \right)^n z^{2n}.$

Problem 4 Sketch the path γ and evaluate the integral $\int_{\gamma} z \, dz$ for the following paths:

(a) $\gamma(t) = e^{it}, t \in [0, \pi],$

(b) $\gamma(t) = \begin{cases} t - 1, & 0 \leq t \leq 2, \\ 1 + i(t - 2), & 2 < t \leq 3, \end{cases}$

(c) $\gamma(t) = \begin{cases} (1 + i)t & 0 \leq t \leq 1, \\ \sqrt{2}e^{\frac{1}{4}i\pi t}, & 1 < t \leq 7, \\ (1 - i)(8 - t), & 7 < t \leq 8, \end{cases}$

(d) $\gamma(t) = \begin{cases} e^{it}, & 0 \leq t \leq \pi, \\ e^{-it}, & \pi < t \leq 2\pi. \end{cases}$

Problem 5 Let $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$. Evaluate the following integrals:

- (a) $\int_{\gamma} 1 \, dz$,
- (b) $\int_{\gamma} |z|^4 \, dz$,
- (c) $\int_{\gamma} (\operatorname{Re} z)^2 \, dz$,
- (d) $\int_{\gamma} (\bar{z})^2 \, dz$,
- (e) $\int_{\gamma} z^{-2}(z^4 - 1) \, dz$,
- (f) $\int_{\gamma} \sin z \, dz$,
- (g) $\int_{\gamma} z^{-1} \left(\bar{z} - \frac{1}{2} \right) \, dz$.

Problem 6 Evaluate the following integrals:

- (a) $\int_{\gamma} \frac{\sin z}{2\pi - z} \, dz$, where $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$,
- (b) $\int_{\gamma} \frac{\sin z}{2\pi - z} \, dz$, where $\gamma(t) = 2e^{it}$, $t \in [0, 2\pi]$,
- (c) $\int_{\gamma} \frac{e^z}{z(z+1)} \, dz$, where $\gamma(t) = 1 + 3e^{it}$, $t \in [0, 2\pi]$,
- (d) $\int_{\gamma} \frac{z^2}{(z+2)(z-1)} \, dz$, where $\gamma(t) = \sqrt{2}e^{it}$, $t \in [0, 2\pi]$,
- (e) $\int_{\gamma} z^n(1-z)^m \, dz$, where $n, m \in \mathbb{Z}$, $m \geq 0$ and $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$,
- (f) $\int_{\gamma} \frac{z^4}{(z-5)^3} \, dz$, where $\gamma(t) = 5 + 2e^{it}$, $t \in [0, 2\pi]$,
- (g) $\int_{\gamma} \frac{e^z}{(z-5)^4} \, dz$, where $\gamma(t) = 5 + 2e^{it}$, $t \in [0, 2\pi]$.

Problem 7 Let f be a holomorphic function in the open disc $D(z_0; R) = \{z \in \mathbb{C} : |z - z_0| < R\}$ with some $R > 0$. Assume that $|f(z)| \leq M$ for all $z \in D(z_0; M)$. Prove that

$$|f^{(n)}(z_0)| \leq \frac{n!M}{R^n}, \quad n \in \mathbb{N}.$$

Problem 8 Let f be holomorphic in the open unit disc $D(0;1) = \{z \in \mathbb{C} : |z| < 1\}$. Prove that if

$$|f(z)| \leq 1 - |z| \quad \text{for all } z \in D(0;1),$$

then $f \equiv 0$ in $D(0;1)$.

Problem 9

(a) Is there a function f that is holomorphic at 1 and satisfies $f\left(\frac{n+1}{n}\right) = f\left(\frac{n-1}{n}\right) = \frac{1}{n^2}$ for all $n \in \mathbb{N}$?

(b) Is there a function f that is holomorphic at 1 and satisfies $f\left(\frac{n+1}{n}\right) = f\left(\frac{n-1}{n}\right) = \frac{1}{n}$ for all $n \in \mathbb{N}$?

Problem 10 Which of the following functions have a removable singularity at the given point z_0 ?

(a) $\frac{1 - \cos z}{z}$ at $z_0 = 0$,

(b) $z \cot z$ at $z_0 = 0$,

(c) $\frac{\cos z}{1 - \sin z}$ at $z_0 = \frac{\pi}{2}$.

Problem 11 Which of the following functions have a pole at the given point z_0 ? If z_0 is a pole, determine its order.

(a) $\sum_{n=0}^{\infty} (-1)^n z^n$ at $z_0 = -1$,

(b) $\frac{\cos z}{\left(\frac{\pi}{2} - z\right)^7}$ at $z_0 = \frac{\pi}{2}$.

Problem 12 Suppose that a function f has a simple pole at $z_0 \in \mathbb{C}$ with residue R . Consider the function

$$g(z) = f(z) - \frac{R}{z - z_0}, \quad z \in \mathbb{C} \setminus \{z_0\}.$$

Show that g has a removable singularity at z_0 .

Problem 13 Calculate residues of given functions at the indicated points:

(a) $\frac{z}{(2 - 3z)(4z + 3)}$ at all singularities,

(b) $\frac{e^{z-1}}{e^z - 1}$ at $z = 0$,

- (c) $\frac{e^{i\pi z}}{16 - z^4}$ at $z = 2$,
- (d) $\frac{\sin z}{1 - 2 \cos z}$ at $z = \frac{\pi}{3}$,
- (e) $\frac{\cos^3 z}{(2\pi - z)^3}$ at $z = 2\pi$,
- (f) $z \tan z$ at $z = \frac{\pi}{2}$,
- (g) $\frac{z^2}{z^4 + 1}$ at $z = \frac{\sqrt{2}}{2}(1 + i)$,
- (h) $\frac{z + 1}{(z^2 + 4)^2}$ at $z = 2i$.

Problem 14 Locate and classify the singularities in $\mathbb{C}$ of the following functions. For those singularities that are poles, calculate the residues.

- (a) $\frac{1}{z^3(z^2 + 1)}$,
- (b) $\frac{1 - e^{iz}}{z^2}$,
- (c) $\frac{e^{iz}}{(z^2 + z + 1)^2}$,
- (d) $\frac{\cot(\pi z)}{z - 1}$,
- (e) $\frac{z \sin z}{\cos z - 1}$,
- (f) $\frac{\sin(2z)}{(z + 1)^3}$,
- (g) $\frac{1}{\sin z}$,
- (h) $\sin\left(\frac{z}{z + 1}\right)$,
- (i) $\sin z \sin \frac{1}{z}$.

Problem 15 Calculate the following integrals using residues:

- (a) $\int_{\gamma} \frac{z^3}{z^4 - 1} dz$, where $\gamma(t) = 2e^{it}$, $t \in [0, 2\pi]$,

- (b) $\int_{\gamma} \frac{1}{z^4 + 1} dz$, where $\gamma(t) = 1 + e^{it}$, $t \in [0, 2\pi]$,
- (c) $\int_{\gamma} z^n e^{\frac{2}{z}} dz$, where $n \in \mathbb{N}$ and $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$,
- (d) $\int_{\gamma_n} \tan(\pi z) dz$, where $n \in \mathbb{N}$ and $\gamma_n(t) = ne^{it}$, $t \in [0, 2\pi]$,
- (e) $\int_{\gamma} \frac{1}{(z-1)^2(z^2+1)} dz$, where $\gamma(t) = 2e^{it}$, $t \in [0, 2\pi]$,
- (f) $\int_{\gamma} \frac{1}{1+e^z} dz$, where $\gamma(t) = 8e^{it}$, $t \in [0, 2\pi]$.

Problem 16 Evaluate the following real integrals using methods of complex analysis:

- (a) $\int_0^{2\pi} \frac{1}{13 - 5 \sin \theta} d\theta$,
- (b) $\int_0^{2\pi} \frac{\cos^2 \theta}{5 + 3 \sin \theta} d\theta$,
- (c) $\int_0^{\pi} \frac{1}{(3 + 2 \cos \theta)^2} d\theta$,
- (d) $\int_{-\pi}^{\pi} \frac{1}{a + b \sin \theta} d\theta$, where $a > b > 0$,
- (e) $\int_{-\infty}^{\infty} \frac{x+4}{(x^2+9)(x^2+2x+2)} dx$,
- (f) $\int_{-\infty}^{\infty} \frac{1}{(x+i)(x^2-2ix-2)} dx$,
- (g) P.V. $\int_{-\infty}^{\infty} \frac{1}{x^3 + 4x + 5} dx$,
- (h) $\int_0^{\infty} \frac{x \sin x}{16x^2 + 9} dx$,
- (i) P.V. $\int_{-\infty}^{\infty} \frac{\cos x}{x^3 + x^2 + 4x + 4} dx$,
- (j) $\int_0^{\infty} \frac{\cos x}{(x^2 + 4)^2} dx$,
- (k) P.V. $\int_{-\infty}^{\infty} \frac{x \sin x}{4 - x^2} dx$,
- (l) P.V. $\int_{-\infty}^{\infty} \frac{\sin x}{(x-1)(x+2)} dx$.

Problem 17 Expand each of the functions in a Laurent series.

(a) $\frac{\cos z}{z^2}$ about $z_0 = 0$,

(b) $\frac{3}{z^2 - z - 2}$ in the annulus $\{z \in \mathbb{C} : 1 < |z| < 2\}$,

(c) $e^{\frac{1}{z-1}}$ about $z_0 = 1$.

Problem 18 How many solutions does the equation

$$z^5 + 15z + 1 = 0$$

has in the annulus $\{z \in \mathbb{C} : \frac{3}{2} < |z| < 2\}$?

Problem 19 How many solutions does the equation

$$z^4 + z + 1 = 0$$

has in the upper half-plane $\{z \in \mathbb{C} : \operatorname{Re} z > 0\}$?

Solutions

Problem 1. $\{0\}$.

Problem 3. (a) $\frac{1}{2}$, (b) $\frac{1}{\sqrt{2}}$, (c) 2, (d) 1, (e) $\frac{1}{2}$, (f) $\sqrt{2}$.

Problem 4. (a) $-2i$, (b) $\frac{1}{2} + i$, (c) 0, (d) 0.

Problem 5. (a) 0, (b) 0, (c) 0, (d) 0, (e) 0, (f) 0, (g) $-\pi i$.

Problem 6. (a) 0, (b) πi , (c) $2\pi i(1 - e^{-1})$, (d) $\frac{2}{3}\pi i$, (e) $(-1)^n 2\pi i \binom{m}{-n-1}$ if $-m-1 \leq n \leq -1$, and 0 otherwise, (f) $300\pi i$, (g) $\frac{1}{3}\pi i e^5$.

Problem 9. (a) Yes, $f(z) = (z-1)^2$. (b) No.

Problem 10. (a) removable, (b) removable, (c) not removable.

Problem 11. (a) simple pole, (b) pole of order 6.

Problem 13. (a) $-\frac{2}{51}$ and $-\frac{3}{68}$, (b) e^{-1} , (c) $-\frac{1}{32}$, (d) $\frac{1}{2}$, (e) 1, (f) $-\frac{\pi}{2}$, (g) $\frac{\sqrt{2}}{8}(1-i)$, (h) $-\frac{1}{32}i$.

Problem 14. (a) Simple poles at $z = \pm i$ with residues $\frac{1}{2}$. A triple pole at $z = 0$ with residue -1 . (b) A simple pole at $z = 0$ with residue $-i$. (c) A double pole at $z = -\frac{1}{2}(1 - i\sqrt{3})$ with residue $-\frac{1}{9}(2\sqrt{3} + 3)ie^{-\frac{1}{2}(\sqrt{3}+i)}$. A double pole at $z = -\frac{1}{2}(1 + i\sqrt{3})$ with residue $\frac{1}{9}(2\sqrt{3} + 3)ie^{\frac{1}{2}(\sqrt{3}-i)}$. (d) Simple poles at $z = k$, where $k \in \mathbb{Z} \setminus \{1\}$ with residues $\frac{1}{(k-1)\pi}$. A double pole at $z = 1$ with residue 0. (e) A removable singularity at $z = 0$. Simple poles at $z = 2\pi k$, where $k \in \mathbb{Z} \setminus \{0\}$ with residues $-4\pi k$. (f) A triple pole at $z = -1$ with residue $2\sin 2$. (g) Simple poles at $z = k\pi$, where $k \in \mathbb{Z}$, with residues $(-1)^k$. (h) An essential singularity at $z = -1$. (i) An essential singularity at $z = 0$.

Problem 15. (a) $2\pi i$, (b) $-\frac{\pi i}{\sqrt{2}}$, (c) $2\pi i \frac{2^{n+1}}{(n+1)!}$, (d) $-4ni$, (e) 0, (f) $-4\pi i$.

Problem 16. (a) $\frac{\pi}{6}$, (b) $\frac{2}{9}\pi$, (c) $\frac{3}{5\sqrt{5}}\pi$, (d) $\frac{2\pi}{\sqrt{a^2-b^2}}$, (e) $\frac{13}{51}\pi$, (f) $\frac{2}{5}\pi i$, (g) $\frac{3}{7\sqrt{19}}\pi$, (h) $\frac{1}{32}\pi e^{-\frac{3}{4}}$, (i) $\frac{1}{10}\pi(2\sin 1 + e^{-2})$, (j) $\frac{3}{32}\pi e^{-2}$, (k) $-\pi \cos 2$, (l) $\frac{1}{3}\pi(\cos 1 - \cos 2)$.

Problem 17. (a) $\sum_{n=0}^{\infty} (-1)^n \frac{z^{2n-2}}{(2n)!}$, (b) $-\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} + \sum_{n=1}^{\infty} (-1)^n z^{-n}$, (c) $\sum_{n=0}^{\infty} \frac{1}{n!(z-1)^n}$.

Problem 18. 4.

Problem 19. 2.